HR Analytics Dashboard

Project Documentation

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Tool	Tableau Desktop
Data Source	Excel Flat File (HR Data.xlsx)
Domain	Human Resources Analytics
Dashboard	https://public.tableau.com/app/profile/shivani.jannaikode1044/viz/HRAnalyticsDashboard_17805346818590/HRAnalyticsDashboard

1. Problem Statement

Employee attrition is one of the biggest challenges organizations face because replacing employees is expensive and impacts productivity. HR teams often struggle to answer key workforce questions without centralized analytics.

Questions this dashboard answers:

Question	Answered By
Which departments are losing the most employees?	Department-wise Attrition pie chart
Which age groups are most likely to leave?	Age group histogram + gender/age donuts
Does education background influence attrition?	Education Field attrition bar chart
How satisfied are employees in different roles?	Job Satisfaction heatmap
Are there gender-based attrition patterns?	Attrition By Gender chart
What is the overall workforce status?	KPI cards at the top of the dashboard

Goal

Develop an interactive Tableau dashboard that enables HR managers and business leaders to monitor workforce metrics, track employee attrition, analyze workforce demographics, understand employee satisfaction patterns, identify groups with higher turnover risk, and support retention planning.

2. Data Source & Preparation

The data is an Excel flat file (HR Data.xlsx) containing 1,470 employee records with 35+ fields. It was cleaned in Excel before connecting to Tableau.

Data Fields

Category	Fields
Demographics	Age, Gender, Marital Status, Education, Education Field
Job Information	Department, Job Role, Job Level, Business Travel, Over Time
Compensation	Monthly Income, Daily Rate, Hourly Rate, Monthly Rate, Percent Salary Hike, Stock Option Level
Satisfaction	Job Satisfaction, Environment Satisfaction, Job Involvement, Relationship Satisfaction, Work Life Balance
Work History	Total Working Years, Years At Company, Years In Current Role, Years Since Last Promotion, Years With Curr Manager
Target Variable	Attrition (Yes / No) — the primary field driving all attrition analysis

Data Cleaning Steps (done in Excel)

- Removed duplicate rows to ensure each employee appears exactly once.
- Checked and handled null/missing values across all columns.
- Verified data types — numeric fields (Age, Income) vs text fields (Attrition, Department).
- Standardized categorical values — Attrition field consistently formatted as 'Yes' or 'No'.
- Removed columns not needed for the analysis (e.g. Over18, StandardHours).
- Validated that total Employee Count matched unique records after cleanup.

3. Calculated Fields

Three calculated fields were created inside Tableau to derive the core metrics used across the dashboard.

Attrition Count

```
IF [Attrition] = "Yes" THEN 1 ELSE 0 END
```

The raw Attrition field contains text values ('Yes' or 'No'). Tableau cannot aggregate text, so this field converts each 'Yes' into a 1 and 'No' into a 0. This allows Tableau to sum attrition across any dimension — by department, gender, age group, and so on.

Result: SUM = 237 (total employees who left)

Active Employees

```
SUM([Employee Count]) - SUM([Attrition Count])
```

Calculates the number of currently retained employees by subtracting total attrition from total headcount. This metric updates automatically whenever a filter is applied on the dashboard, giving the active count for any selected segment.

Result: 1,470 – 237 = 1,233 active employees

Attrition Rate

```
SUM([Attrition Count]) / SUM([Employee Count])
```

Calculates the percentage of the workforce that has left. Formatted as a percentage in Tableau. This is the headline KPI — the primary metric HR managers and executives use to assess workforce retention health.

Result: 237 ÷ 1,470 = 16.12%

4. Parameter — Bin Size

A Tableau parameter is a user-controlled variable that dynamically changes how data is displayed without modifying the underlying dataset. The Bin Size parameter controls the age group intervals in the employee age histogram.

Property	Value
Name	Bin Size
Data Type	Float
Minimum	2
Maximum	10
Step Size	1
Default Value	3
Controls	Age bin width in the 'No of Employees by Age Group' histogram

How Bin Size affects the chart:

Bin Size	Age Groups
3	18–21, 21–24, 24–27, 27–30, 30–33 ...
5	18–23, 23–28, 28–33, 33–38 ...
10	18–28, 28–38, 38–48, 48–58 ...

5. Dashboard Visualizations

The dashboard contains seven visualizations, each built on a separate Tableau worksheet and assembled into the final dashboard layout.

5.1 KPI Cards

Business Question: What is the overall workforce status?

A text/number view (Tableau measure values on columns) displaying five headline metrics: Employee Count (1,470), Attrition Count (237), Attrition Rate (16.12%), Active Employees (1,233), and Average Age (37). These update when the global Education filter is applied.

5.2 Attrition By Gender

Business Question: Is attrition higher among males or females?

A horizontal bar chart with Gender on rows and SUM(Attrition Count) on columns. Male employees account for 150 departures (63.3%) vs 87 female (36.7%). This chart is also an action filter source — clicking a bar cross-filters all other charts on the dashboard.

5.3 Department-wise Attrition

Business Question: Which department loses the most employees?

A pie chart showing attrition split across three departments: Sales 133 (56.12%), R&D; 92 (38.82%), HR 12 (5.06%). Sales accounts for more than half of all attrition. Clicking a slice cross-filters the entire dashboard.

5.4 Number of Employees by Age Group

Business Question: What does the workforce age profile look like?

A vertical bar chart (histogram) using the Age (bin) dimension on columns and SUM(Employee Count) on rows. The Bin Size parameter dynamically controls the width of each age bucket. At Bin Size = 3, the 33–36 age range peaks at 213 employees, with the workforce heavily concentrated in the 27–42 range.

5.5 Job Satisfaction Rating

Business Question: Which job roles have lower satisfaction scores?

A highlight table (heatmap) with Job Satisfaction levels (1–4) on columns and Job Role on rows. Cell color intensity represents employee count — darker means more employees at that satisfaction level. Sales Executives have the highest count at satisfaction level 4 (112 employees).

5.6 Education Field-wise Attrition

Business Question: Which educational backgrounds show higher turnover?

A horizontal bar chart with Education Field on rows and SUM(Attrition Count) on columns. Life Sciences leads with 89, followed by Medical (63), Marketing (35), Technical Degree (32), Other (11), and Human Resources (7).

5.7 Attrition Rate by Gender for Different Age Groups

Business Question: Which demographic segment is at the highest attrition risk?

A row of five donut charts — one per age band (Under 25, 25–34, 35–44, 45–54, Over 55) — split by gender. The central number shows total attrition for that age group. The 25–34 band is the highest with 112 events. Attrition drops significantly for employees aged 45 and above.

6. Dashboard Actions & Interactivity

The dashboard uses Tableau Action Filters to enable cross-chart interactivity. When a user clicks any element in a source chart, all other charts automatically filter to match the selected dimension. Clicking the same element again deselects it and restores the full view.

Action Name	Source Chart	Filter Field	Effect
Action (Department)	Department Pie Chart	Department	All charts update to show only the selected department's employees
Action (Gender)	Gender Bar Chart	Gender	All charts update to show Male or Female employees only
Action (Education Field)	Education Field Bars	Education Field	All charts update to show employees with the selected education background

Global Filter

An Education filter is applied to all sheets simultaneously, allowing the viewer to isolate the entire dashboard by education level: Associates Degree, Bachelor's Degree, Doctoral Degree, High School, or Master's Degree.

7. Key Insights

Insight 1 — Attrition Rate is 16.12%

Approximately 16 out of every 100 employees have left the organization. This exceeds the generally healthy benchmark of 10–12% annually, indicating that retention efforts are needed across the organization.

Insight 2 — Sales Has the Highest Attrition (56.12%)

Sales accounts for 133 of 237 total departures — more than half of all attrition despite not being the largest department. This may reflect high-pressure targets, performance quotas, or compensation structure issues. HR should investigate further through exit interviews focused on Sales employees.

Insight 3 — Employees Aged 25–34 Are the Highest Attrition Group (112 events)

The 25–34 age band accounts for nearly half of total attrition. This segment typically has enough experience to be marketable externally and is often seeking faster career progression, better compensation, or more meaningful work. Retention efforts targeted at this cohort could have a significant business impact.

Insight 4 — Life Sciences & Medical Backgrounds Show the Highest Turnover

Employees with Life Sciences (89) and Medical (63) educational backgrounds account for 64% of education-related attrition. These fields have strong external demand from pharmaceutical, biotech, and healthcare sectors, giving these employees abundant outside opportunities.

Insight 5 — Male Attrition Exceeds Female Attrition

150 male vs 87 female employees left the organization. While this may partly reflect the overall workforce gender composition, it warrants a deeper analysis of role distribution, work-life balance offerings, and career growth perception by gender.

8. Skills Demonstrated

Skill Area	Details
Data Preparation	Excel data cleaning, null handling, data type validation, column normalization
Tableau Fundamentals	Connecting Excel to Tableau, dimension vs measure management, field type inference
Calculated Fields	IF/THEN/ELSE logic, SUM aggregation, derived metric creation
Parameters	Creating numeric parameters, binding parameters to bins for dynamic segmentation
Chart Design	KPI text views, horizontal bars, pie charts, histograms, highlight tables (heatmaps), donut charts
Dashboard Design	Multi-sheet assembly, layout containers, filter controls, color theming
Interactivity	Tableau action filters, global filter application, parameter controls on dashboard
Business Analysis	Translating HR business questions into chart specifications, insight derivation
Data Storytelling	Executive-level readability, visual hierarchy, actionable insight communication

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